Policy gradient methods

Lecture 2: baselines, actor-critic, TD, GAE and PPO

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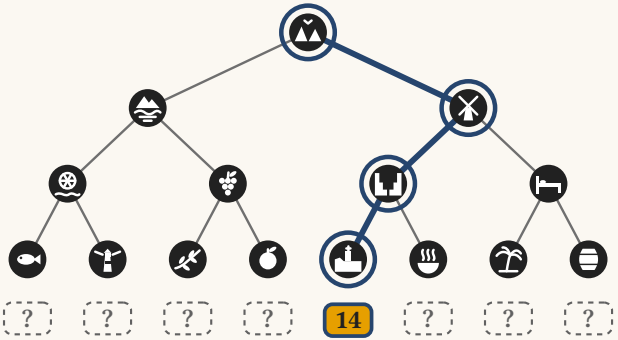
20 October 2026

5

Part 5

REINFORCE with a baseline

G_t alone is not very informative

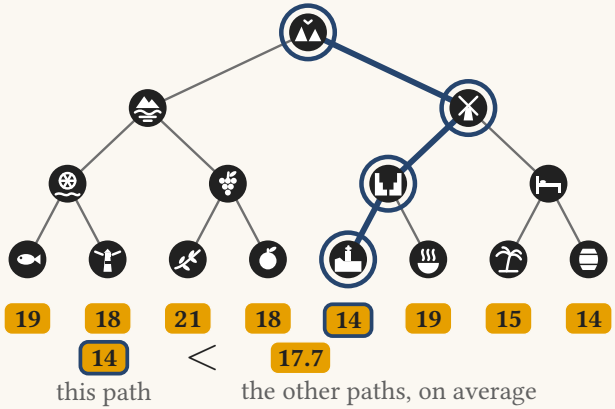


1 One day on the outlined path paid $G_1 = 14$ coins, luck included.

2 Is this path good? Is it bad?

3 It depends on what the other paths yield.

If the other paths pay more, this path is poor

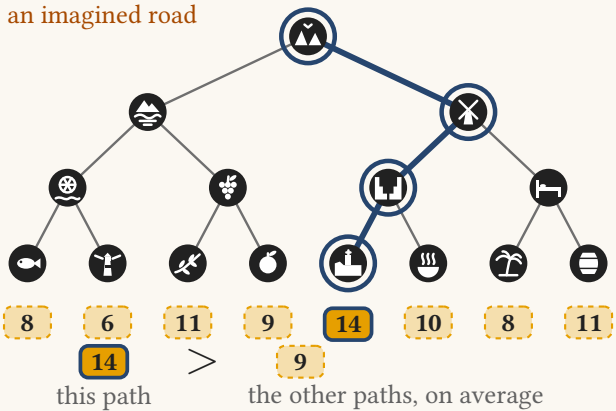


1 The gold numbers are what a day on each path pays on average.

2 The other 7 paths average 17.7 coins, more than the 14 of this day.

3 Next to them, this path is poor.

If the other paths pay less, this path is good



1 On an imagined road the other paths would pay the dashed numbers.

2 The other 7 paths average 9 coins, less than the 14 of this day.

3 Next to them, the same path is good. The multiplier should include what the other paths yield.

The multiplier compares a path with the average

$$\cancel{\Psi_t = G_t} \longrightarrow \Psi_t = G_t - \boxed{V(s_t)}$$

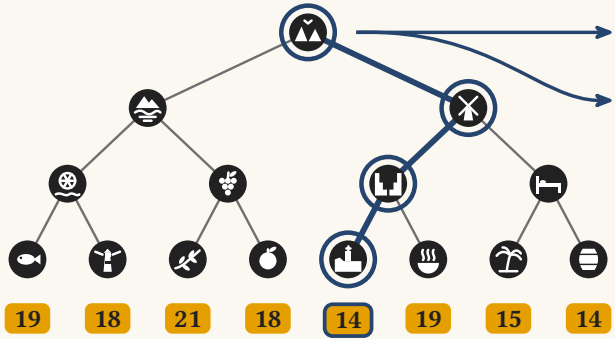
the average of G_t over all the paths from s_t

1 Part 4's multiplier is $\Psi_t = G_t$. Coins alone do not say whether a path is good.

2 The new multiplier subtracts $V(s_t)$. A number subtracted like this is a baseline.

3 $V(s_t)$ averages G_t over all the paths that π_w can take from s_t, each weighted by its chance, the path itself included.

Example: the uniform policy gives $V(s_1) = 17.25$



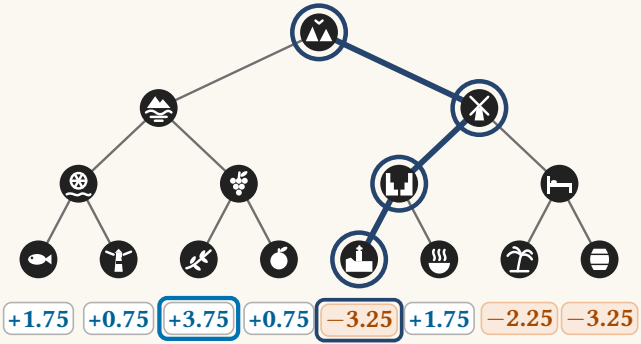
$G_1 = 14$

$V(s_1)$: the average of G_1 over all 8 paths

1. The uniform policy: every road has chance 1/2, so each path has chance 1/8.
2. The outlined path is one of the eight. The other 7 average 17.7.

$$V(s_1) = (19 + 18 + 21 + 18 + 14 + 19 + 15 + 14) \,/\, 8 = 17.25$$

The best path gets the largest positive multiplier

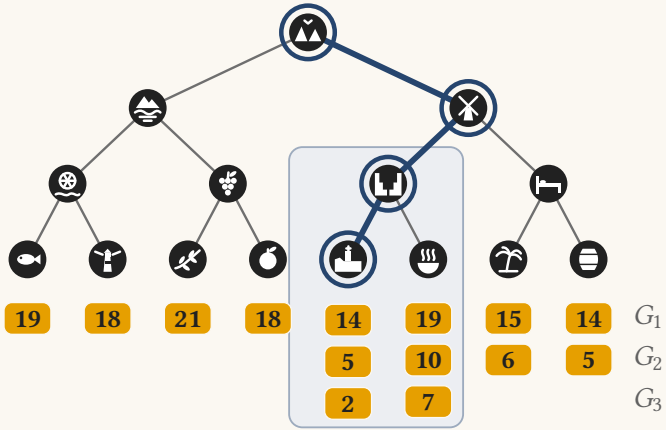


1 For each path, $\Psi_1 = G_1 - V(s_1)$. For the outlined path, $14 - 17.25 = -3.25$.

2 The best path, 21 coins, gets the largest multiplier, +3.75.

3 The three paths below the average get a negative multiplier.

$V(s_t)$ depends on where Mara stands

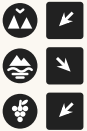


1 $V(s_t)$ averages G_t over the paths still ahead. At step 1 all 8 are: $V(s_1) = 17.25$.

2 At step 2 Mara stands at the market and 4 paths remain: $V(s_2) = 6.5$.

3 At step 3 she stands at the canyon and 2 remain: $V(s_3) = 4.5$.

The poor day now pushes its roads down

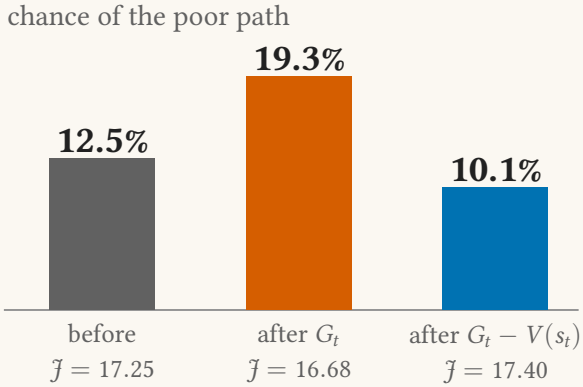
		G_t		$V(s_t)$		Ψ_t
poor day		14	−	17.25	=	−3.25
		4	−	6.5	=	−2.5
		2	−	4.5	=	−2.5
good day		21	−	17.25	=	+3.75
		14	−	12	=	+2
		6	−	5.5	=	+0.5

1 Part 4 pushed every road by G_t: even the poor day pushes its three roads up.

2 $V(s_t)$ is the average of G_t from the place where the road was taken.

3 The new push is $G_t - V(s_t)$: the poor day's roads go down, the good day's stay up.

With the baseline the poor path becomes rarer

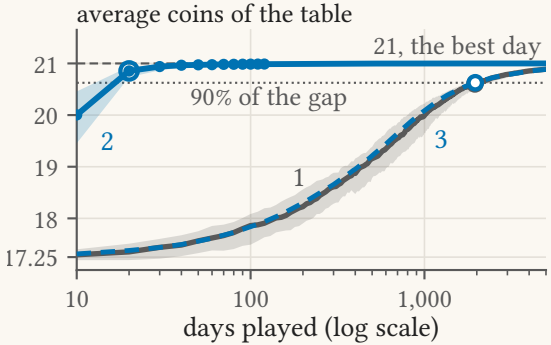


1 From the uniform table, update the knobs with the poor day alone, with step size 0.05.

2 With G_t the poor path becomes likelier and $\mathcal{J}$ falls.

3 With $G_t - V(s_t)$ it becomes rarer and $\mathcal{J}$ rises. One day is a small sample: on average the two updates agree.

REINFORCE with a baseline converges much faster



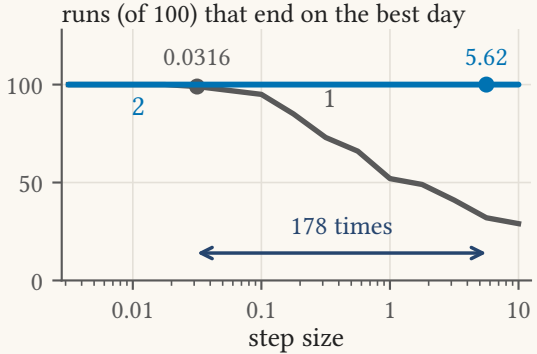
Median of 200 runs, with the middle half. The safe step is the fastest at which 99 of 100 other runs close 90% of the gap.

1 REINFORCE at its safe step, 0.0316, closes 90% of the gap from 17.25 to 21 coins in a median of 1,950 days.

2 With the baseline, at its safe step, 5.62, it takes 20 days: 97.5 times fewer. Here $V(s_t)$ is exact.

3 At REINFORCE's step the same baseline is no faster: 1,960 days.

The baseline allows a much larger step



A run ends on the best day when its likelier roads lead to the 21-coin path.

1 REINFORCE fails more often above 0.0316: at 5.62, 32 of 100 runs end on the best day.

2 With the baseline all 100 do, up to 10, the largest step tried. Its safe step is 178 times larger.

3 At one common step the two take equally long, since their average push is the same. The baseline's push is quieter, and that allows the larger step.

Here $V(s_t)$ is computed from the road itself

```
w = knobs, all 0
repeat:
1   compute V(s) for every place
    play N days with softmax(w)
    push every road of every day
2     as hard as the coins after it
      minus V(s_t)
3   w = w + alpha * average push
```

1 $V(s)$ of every place, from the table and the road: the tree is small, so it is exact.

2 Every road is pushed by $G_t - V(s_t)$ instead of G_t.

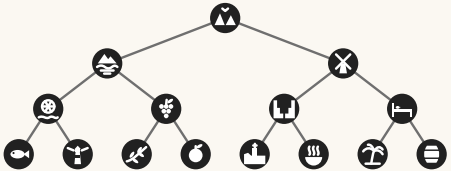
3 The knobs move as before. Every run of the last two frames used this exact V.

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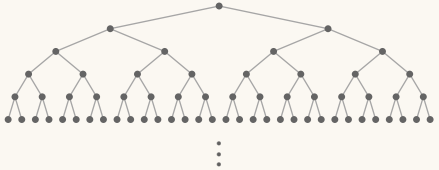
Part 6

Actor-critic

$V(s_t)$ is usually very hard to calculate



Mara's road: eight paths



chess, a language model

1 Part 5's multiplier $\Psi_t = G_t - V(s_t)$ needs $V(s_t)$: on Mara's road an average over eight paths. We listed them.

2 In chess, or with a language model, no one can list the paths: $V(s_t)$ is very hard to calculate.

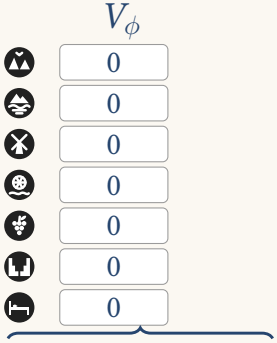
So we estimate it!

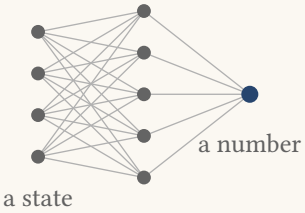
the actor
π_w, the table of knobs

→

the critic
V_ϕ, an estimate of V

A critic is a function that estimates V





a neural network

1 A critic is a function V_ϕ: from a place it returns a number, our estimate of V. Its weights are ϕ.

2 On Mara's road it is a table, one number for each place with a choice. We start it at zeros.

3 In real life a neural network reads the state. Either way $V_\phi(s_t)$ takes the place of $V(s_t)$ in Ψ_t.

Every step of every day is a labelled example



1 Play N days with π_w: here the five days of Part 2.

2 Each step is an example: the place s_t where it starts, and its label, the coins G_t from there to the end of the day.

3 It is a data set: learn to guess G_t from the place s_t.

The critic's loss is the squared gap

$$L(\phi) = \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \tfrac{1}{2} \Big(\underbrace{G_t^{(n)} - V_\phi(s_t^{(n)})}_{\text{the gap}} \Big)^2$$

	the labels of each place	their average
	15, 17, 14, 15, 19	**16.0**
	10	**10.0**
	6, 6, 6, 9	**6.75**
	none	none
	3	**3.0**
	2, 6	**4.0**
	1, 3	**2.0**

1 One example costs half the squared gap between the critic's number and its label.

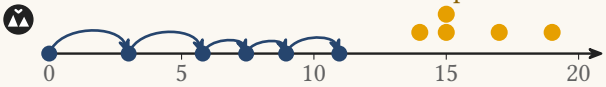
2 Each place has its own part of the loss: its labels only.

3 A part is smallest at the average of its labels: with many days, $V(s_t)$.

A gradient step moves the table towards the labels

$$\phi \leftarrow \phi - \beta \nabla_\phi \tfrac{1}{2}\big(G_t - V_\phi(s_t)\big)^2$$

$+0.2 \times (15 - 0) = 3.0$



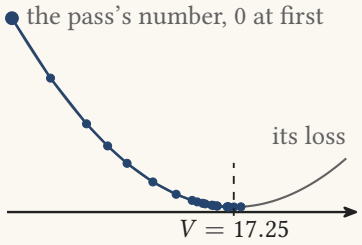
	before	after five days
	0	10.96
	0	2.00
	0	4.14
	0	0.00
	0	0.60
	0	1.52
	0	0.76

1 Only the entry of s_t is in the loss of an example: it moves a fraction β of the way to the label. β is the critic's step.

2 One step per example, in order. Here $\beta = 0.2$. A place that no day visited keeps its number.

Algorithm to estimate the critic

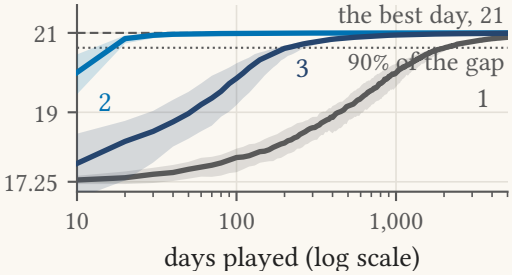
```
w, phi = zeros, zeros
repeat:
  play N days with softmax(w)
  G = coins to come
  Psi = G - phi[s]                    (1)
  w = w + alpha * average push
  for each example, in order:         (2)
    phi -= beta * grad(loss)
```

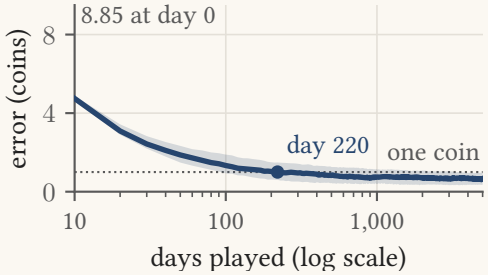


1 The multiplier $G_t - V_\phi(s_t)$ reads the table as it stood before the batch, so it never depends on the roads of the day it multiplies.

2 The critic steps down its own bowl, one example at a time.

A learned critic speeds REINFORCE up





1 REINFORCE closes 90% of the gap in 1,950 days ($\alpha = 0.0316$; median of 200 runs).

2 With the exact V of Part 5: 20 days ($\alpha = 5.62$).

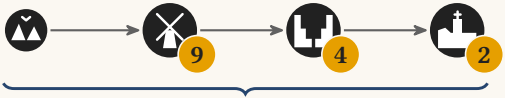
3 A critic learned from zeros: 210 days ($\alpha = 0.316$, $\beta = 0.1$).

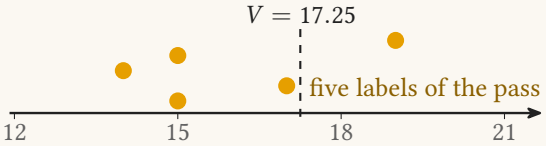
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Part 7

TD learning

Estimating the critic is expensive





1 To label a step we wait until the day ends.

2 A label is a sum of random coins: its spread around V is 2.82 coins.

3 The critic needs many days to average that noise out. Can we find a cheaper way?

The coins to come unfold one step at a time

$$
\begin{aligned}
G_t &= r_t + \gamma\, r_{t+1} + \gamma^2\, r_{t+2} + \cdots + \gamma^{T-t}\, r_T \\
&= r_t + \gamma\, \underbrace{\left(r_{t+1} + \gamma\, r_{t+2} + \cdots + \gamma^{T-t-1}\, r_T\right)}_{G_{t+1}} = r_t + \gamma\, G_{t+1}
\end{aligned}
$$

$G_3 = r_3 = 2$ $\qquad$ $G_2 = r_2 + G_3 = 4 + 2 = 6$ $\qquad$ $G_1 = r_1 + G_2 = 9 + 6 = 15$

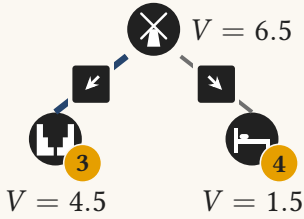
1 G_t adds the coins from step t to the last step T of the day, each one γ times smaller than the one before. On Mara's road $T = 3$ and $\gamma = 1$.

2 Take r_t out: what is left is γ times G_{t+1}.

3 Day 1 of Part 2, from the last step back.

Averaging it gives the Bellman equation

$$V(s_t) = \underbrace{\mathbb{E}\big[\, r_t + \gamma\, V(s_{t+1}) \mid s_t \big]}_{\text{the average over what can follow } s_t} \;\approx\; \underbrace{r_t + \gamma\, V(s_{t+1})}_{\text{one sample of it}}$$



Day 1 took the first road: paid 4 coins, the usual 3 and a lucky one.

1 Average both sides of $G_t = r_t + \gamma G_{t+1}$, given that Mara stands at s_t: over the road, the next place and the luck.

2 At the market: $\frac{1}{2}(3 + 4.5) + \frac{1}{2}(4 + 1.5) = 6.5$.

3 One step gives one sample: $4 + 4.5 = 8.5$.

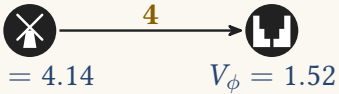
After every step the critic can learn

$$y_t = r_t + \gamma\, V_\phi(s_{t+1})$$

the target: the table's number for the next place stands in for the rest of the day

$$\phi \leftarrow \phi - \beta\, \nabla_\phi \tfrac{1}{2}\Big(\underbrace{y_t}_{\text{held fixed}} - V_\phi(s_t)\Big)^2$$

the gradient goes through $V_\phi(s_t)$ only



$$y = 4 + 1.52 = 5.52$$

$$V_\phi(\text{market}) = 4.14 + 0.2 \times (5.52 - 4.14) = 4.42$$

1 A step of size β with y_t held fixed: a semi-gradient.

2 In the algorithm the label G_t of the critic's step becomes y_t. The multiplier is still $G_t - V_\phi(s_t)$.

8

Part 8

GAE

A third multiplier uses the critic's numbers

the box Ψ_t of the gradient formula holds

Part 4

$$\Psi_t = \boxed{G_t}$$

the coins to come

Parts 5 to 7

$$\Psi_t = \boxed{G_t - V(s_t)}$$

the coins to come minus the critic's number

Part 8

$$\Psi_t = \boxed{\hat{A}_t}$$

built from the critic's numbers

1 The box has held the coins to come, then the coins to come minus the critic's number.

2 A third option, $\hat{A}_t$, starts from the critic's numbers.

3 One number λ sets how much of the day it lets in.

Step 1: a surprise for every step, all at once

$$\delta_t = r_t + \gamma\, V(s_{t+1}) - V(s_t) \quad \text{for } t = 1, \dots, T$$

	step 1	step 2	step 3
			
r_t	9	4	2
$+\,\gamma V(s_{t+1})$	6.5	4.5	0
$-\,V(s_t)$	17.25	6.5	4.5
$=\,\delta_t$	−1.75	+2	−2.5

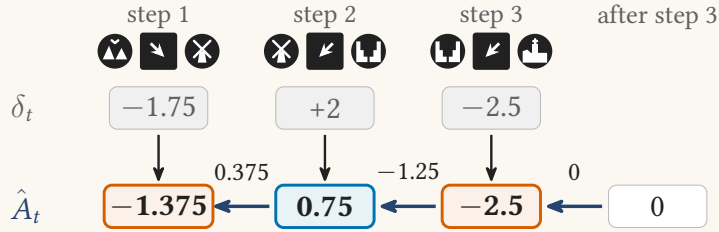
1 The surprise of a step: what Mara got, plus what the critic expects next, minus what it expected here.

2 V is the critic's number for a place; the end is worth 0. Here $T = 3$ and $\gamma = 1$.

3 Positive: the step beat what the critic expected. Negative: it fell short.

Step 2 sweeps from the last step to the first

$$\hat{A}_{T+1} = 0$$
$$\hat{A}_t = \delta_t + \gamma\lambda\,\hat{A}_{t+1} \qquad \text{for } t = T, T-1, \ldots, 1$$

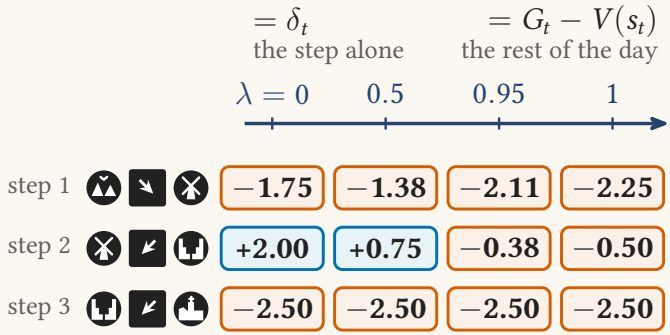


1 The sweep starts after the last step, with $\hat{A}_{T+1} = 0$.

2 Each step adds its surprise to $\gamma\lambda$ times the value of the step after it. Here $\gamma = 1$ and $\lambda = 0.5$.

3 It runs from the last step to the first.

One number λ dials between two multipliers



1 The multiplier is $\Psi_t = \hat{A}_t$, the value the sweep ends with at step t.

2 $\lambda = 0$ gives δ_t. $\lambda = 1$ gives $G_t - V(s_t)$, the multiplier of Part 5: $15 - 17.25 = -2.25$ at step 1.

3 In between, λ sets how much of the rest of the day is let in.

9

Part 9

PPO

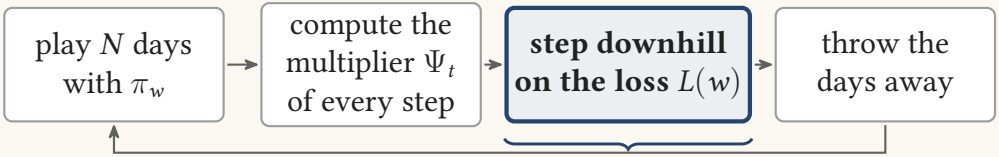
REINFORCE is gradient descent on a loss

$$L(w) \;=\; -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

$$\nabla L(w) \;=\; -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \underbrace{\boxed{\Psi_t^{(n)}} \quad \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})}_{\text{minus the estimate of Part 4}}$$

1. This is Part 4's loss: minus the multiplier times the log chance of the road taken.
2. The multiplier is a plain number: the gradient is minus the estimate of Part 4.
3. A step downhill, $w \leftarrow w - \alpha\,\nabla L(w)$, is a step uphill on $\mathcal{J}$: Part 4's update.

PPO keeps the algorithm and changes the loss

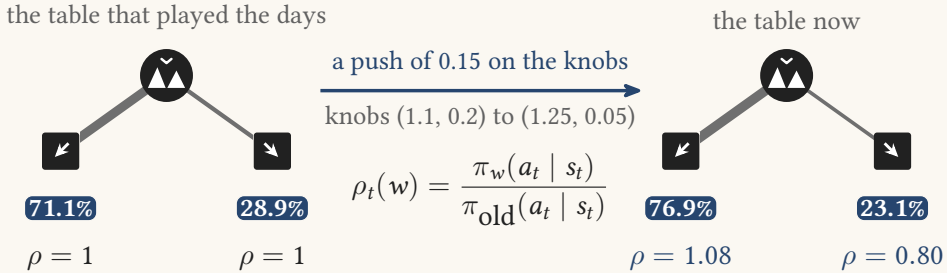


Part 4 $L(w) = -\frac{1}{N} \sum \sum \Psi_t \log \pi_w(a_t \mid s_t)$

Part 9 a new loss, built in the next frames

1. The algorithm of Part 4 stays: play, compute the multiplier, step, discard.
2. The step is gradient descent on a loss: the loss decides what the step does.
3. Autograd differentiates whatever loss we write. PPO writes a new one.

The loss remembers the table that played the days



1 Each step stores the chance its road had in the table that played the day: $\pi_{\text{old}}(a_t \mid s_t)$.

2 After a push the table gives other chances: 76.9% and 23.1% at the pass.

3 The ratio is the new chance over the old one: 1.08 for ↙ and 0.80 for ↘.

The first loss multiplies the ratio by the multiplier

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

replace it by the ratio

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \rho_t^{(n)}(w)$$

1 Part 4's loss has the log chance of the road taken in every term.

2 The new loss uses the ratio instead. At the old table every ratio is 1.

3 So the gradient is Part 4's and the first step is Part 4's step.